

BUSINESS REQUIREMENTS DOCUMENT (BRD)

1. Business Context / Problem Statement

Manufacturing leadership requires a centralized analytics dashboard to monitor operational efficiency, quality performance, downtime reliability, and supplier-related risk across multiple plants. Current reporting lacks integrated visibility into production, quality, and operational drivers.

- Why efficiency fluctuates
- Where losses are occurring
- Which machines or suppliers are underperforming
- How downtime and quality issues impact profitability

2. Goal of the Dashboard

The dashboard must:

- Monitor operational efficiency against production plans.
- Identify quality loss drivers and scrap patterns.
- Detect downtime bottlenecks and machine-level risks.
- Evaluate supplier performance impact.
- Enable plant-level comparative performance analysis.

3. Target Users / Stakeholder Personals

User Personal	Primary Need	Expected Usage
Plant Manager	Monitor plant efficiency & downtime	Daily review
Operations Head	Cross-plant comparison	Weekly strategy
Quality Manager	Identify scrap & QA risks	Root cause analysis
Supply Chain Manager	Evaluate supplier risk	Performance evaluation

4. Core Business Questions

1. Are plants meeting production targets?
 2. What are the primary causes of scrap and quality loss?
 3. Which machines contribute most to downtime?
 4. How does downtime impact yield?
 5. Which suppliers present material risk?
-

5. Product-Style KPIs

KPI	Definition
Plan Attainment %	$(\text{Actual Output} / \text{Planned Output}) \times 100$
Yield %	$(\text{Good Units} / \text{Total Production Units}) \times 100$
Scrap Rate %	$(\text{Scrap Units} / \text{Total Units}) \times 100$
Downtime %	$(\text{Downtime Minutes} / \text{Planned Runtime}) \times 100$
Energy per Good Unit	Energy Used / Good Units
Water per Good Unit	Water Used / Good Units

6. Scope

In Scope:

- Operational performance monitoring
- Production & Throughput
- Quality & scrap analysis
- Downtime reliability tracking
- Supplier risk overview

Out of Scope:

- Financial profitability modeling
 - Workforce productivity analysis
-

7. Success Criteria

- Clear visibility into operational KPIs.
- Ability to identify top 3 loss drivers within 1 minute.
- Plant comparison without external reports.
- Executives able to make data-backed decisions.